

SINGAPORE UNIVERSITY OF
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50.021 - Artificial Intelligence

Final Report

By Group 1 – Artificially Intelligent

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GitHub Repository Link: <https://github.com/chienshyong/artificially-intelligent>

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Introduction

As online platforms become central to public discourse, understanding the tone and nature of conversations taking place is more important than ever. With harmful speech, such as racism, cyberbullying, and other targeted abuse, growing in scale and complexity, there is a pressing need for tools that can automatically identify and classify such content. This project explores the application of sentiment analysis enhanced with multi-class labelling to tackle this challenge.

Following the theme of “AI for Good”, we aim to train an AI model to distinguish between various forms of harmful speech, supporting data-driven efforts in digital safety, moderation, and social research.

Cyberbullying on X

Social media platforms, particularly X (formerly known as Twitter), has significantly amplified incidents of cyberbullying. X was identified as the second most common platform for online harassment, with 27% of surveyed individuals reporting experiences of cyberbullying on the platform. (Sternier 2017) The problem has only gotten worse with the downsizing of the moderation team at X.

95% of social media content creators have encountered harassment or hate during their careers. Cyberbullying increases suicidal thoughts by 14.5% and suicide attempts by 8.7% among adolescents.

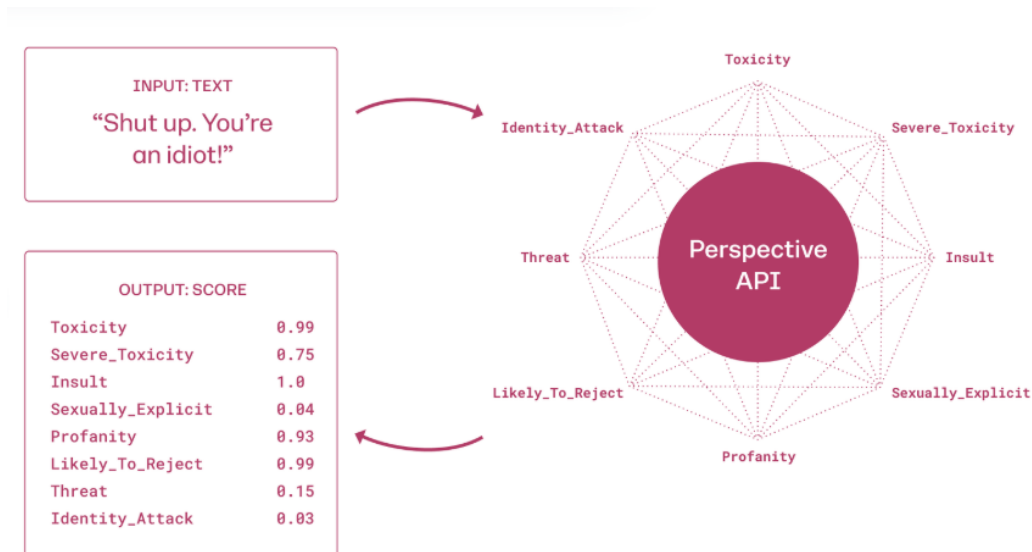
Problem Statement

This study seeks to address the challenge by developing an AI classification model that leverages advanced natural language processing (NLP) techniques to detect and categorize cyberbullying within tweets.

The model trained from our study, achieving an accuracy of 88%, outperforms the baseline Google Jigsaw API with an accuracy of 76% by training specifically for cyberbullying on X. It can assist platform moderators in making more informed, timely decisions that promote healthier online interactions.

Background Research

One popular existing solution for identifying and quantifying toxicity in online comments is Jigsaw's Perspective API. It is widely used for moderation by platforms like Reddit, YouTube, and Wikipedia to moderate comments. It is available to the public so it can be used as a baseline standard for comparing our model.



The API provides a score between 0 (not toxic) and 1 (highly toxic). Websites can use threshold-based filtering to detect toxic comments above their tolerance level (e.g., flagging comments with scores above 0.7).

The training methodology is also public at <https://developers.perspectiveapi.com/s/about-the-api-model-cards>. It uses training data of millions of comments from a variety of sources, labelled by 3-10 raters who speak the relevant language from 0 to 1 on different attributes (toxicity, insult, profanity, threat)

The architecture used by Jigsaw is also public: finetuned Bidirectional Encoder Representations from Transformers (BERT) models. BERT is based on the Transformer encoder architecture, which uses self-attention mechanisms to understand the context of a word in relation to all other words in a sentence. BERT is bidirectional, meaning it considers both the left and right context of a word simultaneously during training, leading to a deeper understanding of language. BERT is initially trained on a large corpus and can be fine-tuned with labelled data for specific tasks.

Model Proposal

To develop an effective AI-powered sentiment analysis system that contributes to positive social impact, we propose utilising a diverse ensemble of models, specifically Efficiently Learning an Encoder that Classifies Token Replacements Accurately (ELECTRA), Decoding-enhanced BERT with disentangled attention (DeBERTa), Robustly Optimized BERT Pretraining Approach (RoBERTa), Random Forest and a stacking of ELECTRA, DeBERTa and the Random Forest. Each model is chosen for its unique strengths in capturing sentiment in complex, real-world text, particularly from social media and user-generated platforms.

ELECTRA is a transformer-based model designed to increase training efficiency and performance. Unlike traditional masked language models such as BERT that learn by predicting missing words, ELECTRA trains a discriminator to detect whether a token in a sentence has been replaced by a generator. This method, known as replaced token detection, allows ELECTRA to leverage all input tokens during training rather than only the masked ones, thereby improving sample efficiency. ELECTRA's lightweight architecture and efficient pre-training make it highly suitable for low-latency applications, offering high classification accuracy with reduced computational cost. In the context of sentiment analysis, this makes ELECTRA an excellent choice for real-time systems where both performance and speed are essential.

RoBERTa (Robustly Optimised BERT Pretraining Approach) refines the original BERT model by removing the Next Sentence Prediction (NSP) task, applying dynamic masking during training, increasing the training batch size, and training over larger datasets for longer durations. These enhancements allow RoBERTa to learn more robust language representations, leading to consistently high performance across a wide range of Natural Language Processing (NLP) benchmarks. RoBERTa is especially effective at capturing global context and semantic coherence in longer texts. In our ensemble, RoBERTa serves as a robust generalist, contributing reliable sentiment classification across varying text lengths and structures, including long-form opinion pieces and threaded discussions.

DeBERTa builds upon the BERT and RoBERTa frameworks by introducing two key innovations: disentangled attention and an enhanced mask decoder. In contrast to standard models that combine positional and content information into a single representation, DeBERTa separates these two components. This disentanglement allows the model to more accurately represent word dependencies and syntactic relationships, especially in complex or ambiguous language. The enhanced mask decoder further improves the model's ability to predict contextually masked tokens. As a result, DeBERTa demonstrates superior performance in tasks requiring fine-grained understanding, such as

detecting implicit sentiment or distinguishing subtle differences in tone. This makes it highly valuable for analysing text containing irony, cultural context, or layered emotional expressions.

Random Forest is an ensemble-based machine learning algorithm that constructs multiple decision trees during training and outputs the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. It operates by sampling data and features randomly to build diverse trees, reducing overfitting and improving generalisation. Though not based on deep learning, Random Forest is robust, interpretable, and effective for tabular or structured data. In the context of sentiment or cyberbullying detection, it can be used with feature-engineered inputs such as TF-IDF vectors or sentiment scores. Within this ensemble, Random Forest provides a reliable, non-neural perspective that complements the deep transformer models by offering consistent predictions based on interpretable decision boundaries.

To exploit the complementary strengths of these models, we implement a hybrid ensemble strategy using stacking.

Stacking is a two-layer ensemble approach. In the first layer, we independently train each model, ELECTRA, DeBERTa and Random Forest, on the sentiment classification task. Each model generates its predictions in the form of either sentiment labels or class probabilities. These predictions are then used as features for a meta-learner, also known as a blender model. The meta-learner is trained on a validation set to learn the optimal way to combine the outputs of the base models. The key advantage of stacking is that it allows each model to specialise in different aspects of the data, resulting in an overall classifier that is more robust and generalisable than any single model.

Training

Dataset

For this project, we selected a dataset comprising 47,000 labelled tweets, originally compiled by the Institute of Electrical and Electronics Engineers (IEEE) for their research into Graph Convolutional Network methods, as detailed in their publication: [SOSNet: A Graph Convolutional Network Approach to Fine-Grained Cyberbullying Detection | IEEE Conference Publication | IEEE Xplore](#)

The dataset is publicly available on Kaggle ([Cyberbullying Classification](#)) under the title Cyberbullying Classification. It consists of tweets that have been manually labelled into six

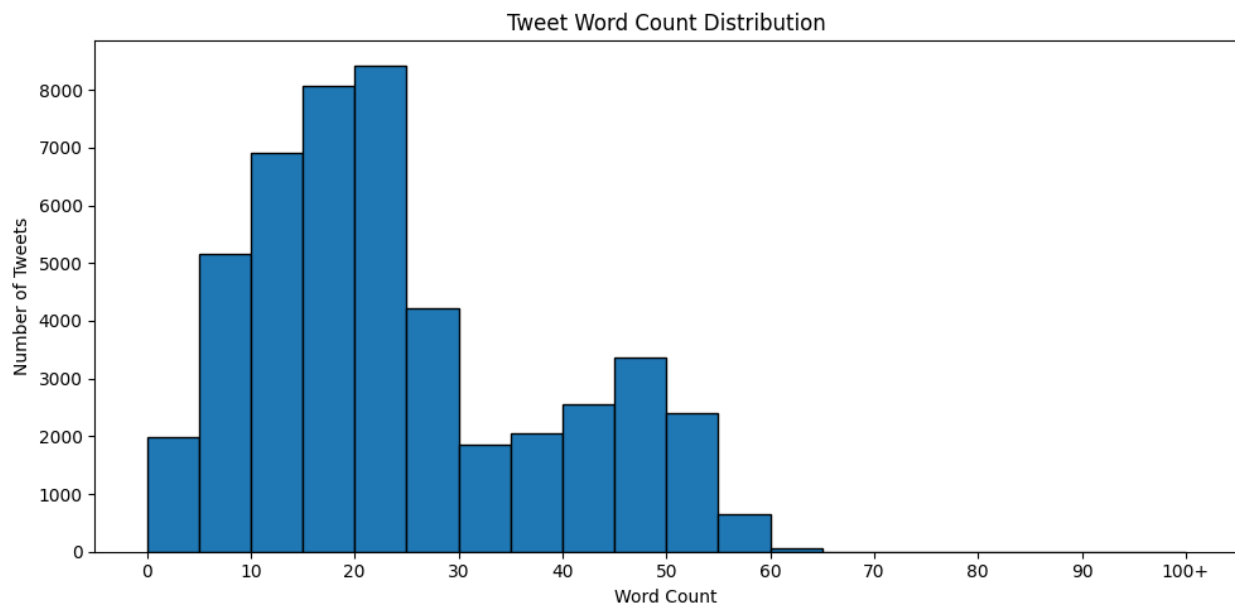
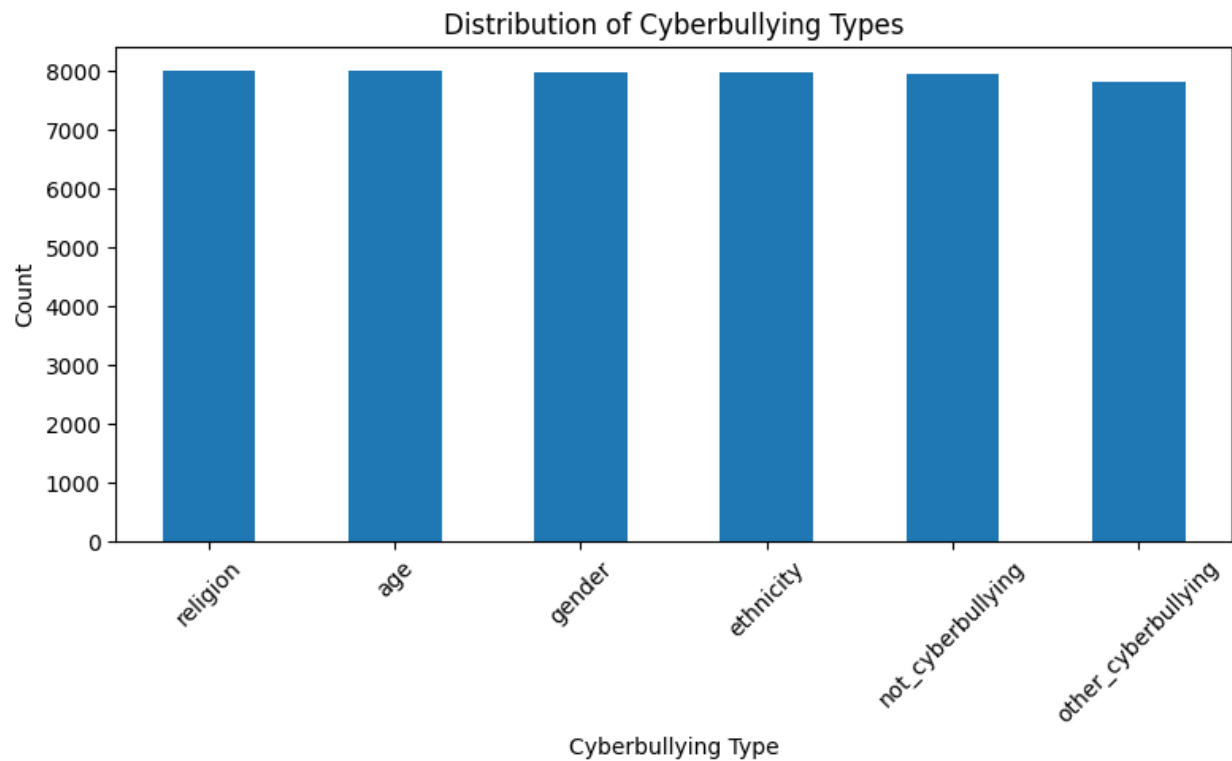
distinct classes based on the type of cyberbullying: ‘age’, ‘ethnicity’, ‘gender’, ‘religion’, ‘other_cyberbullying’ and ‘not_cyberbullying’.

Importantly, the dataset is balanced, with an equal number of samples in each class. This ensures that the model can be trained effectively without inherent bias towards any specific category, supporting more accurate and fair classification outcomes.

Sample Data

tweet_text	cyberbullying_type
The contact form on the site was bugged. Turns out that if you use a 'j' in the email subject prefix, it breaks everything. silly wordpress.	not_cyberbullying
RT @nowlinn Not sexist but I hate female sports announcers	gender
@BilalGhumman @OdinialInvictus @IsraeliRegime That is why we are going secular, microbrain. And Islam is as disgusting as ever.	religion
Lol awww. Bullied too much in high school, girls? Try not to hate the world so much	age
@MsLisaRayy muhaha f u an yo fort lil n a	ethnicity
RT @josezarate620: hello twitter. u ugly as hell tonight.	other_cyberbullying

Data Visualisation



The chart above visualizes the dataset by word count. On Twitter, some users are limited to 140 characters per tweet, while others have access to 280 characters. This leads to an interesting histogram where there are two peaks of word counts: one around 20 words, and one around 45 words.

	age	ethnicity	gender	not_cyberbullying	other_cyberbullying	religion
0	school (0.686)	n[REDACTED] (0.590)	r[REDACTED]e (0.547)	mkr (0.684)	rt (0.424)	muslims (0.443)
1	high (0.420)	f[REDACTED]k (0.443)	gay (0.493)	http (0.394)	http (0.401)	muslim (0.376)
2	bullied (0.363)	dumb (0.442)	jokes (0.435)	rt (0.290)	just (0.270)	idiot (0.277)
3	bully (0.228)	n[REDACTED]s (0.207)	joke (0.246)	bullying (0.266)	bully (0.263)	christian (0.241)
4	girl (0.195)	ass (0.193)	rt (0.157)	just (0.164)	https (0.262)	islamic (0.230)
5	girls (0.194)	tayyoung_ (0.183)	people (0.128)	bully (0.151)	like (0.225)	idiots (0.210)
6	like (0.153)	black (0.179)	sexist (0.124)	like (0.147)	don (0.210)	radical (0.195)
7	bullies (0.135)	white (0.130)	just (0.121)	don (0.127)	people (0.204)	like (0.194)
8	just (0.090)	rt (0.104)	like (0.117)	school (0.126)	f[REDACTED]king (0.176)	islam (0.193)
9	people (0.077)	people (0.100)	women (0.115)	amp (0.104)	mkr (0.157)	terrorism (0.182)

We can use TF-IDF to find the significant words for each class while down-weighting common terms across all classes (sensitive words have been censored).

Preprocessing

The data that we wish to train needs to be pre-processed to ensure that the training data is as accurate as possible. Of all data, text is the most unstructured form, which means that we require multiple steps to help convert noise from high dimensional features to the low dimensional space to obtain as much accurate information as possible from the text. As we are dealing with tweets and sentiment analysis, we need to be specific with the steps that we take to retain as much of the meaning of the text as possible.

We can split the process into a few main categories. Negation, Tokenization, and Lemmatization. Throughout the categories, we also carry out normalization processes to fix different words which look different due to casing or are written in different ways despite possessing similar meanings. An example of this is to change casings.

Negation Handling

Apostrophes connecting words are used everywhere, especially in public reviews. To maintain uniform structure, it is recommended they should be converted into standard lexicons. The text will then follow the rules of context free grammar and helps avoids any word-sense disambiguation.

We have used an edited version of a negation function by Utkarsh Lal (2022), which detects when a negation word is encountered. A set of cognitive synonyms called synsets are generated for the word next to the negation. These synsets are interlinked by conceptual semantic and lexical relations to each other in a lexical database called WordNet, which is part of the NLTK Python library. After the synsets are created, the method checks if an antonym of that word exists in the WordNet. If an antonym doesn't exist, it implies that the word is either a verb or an entity that has no opposite word available in the WordNet. So, it

is left as it is, and the context of the sentence is preserved. But if an antonym exists in the lexical database, then a list of antonyms is created.

Next, the function checks the dissimilarity coefficient of the list of antonyms and chooses the one with the highest dissimilarity. Then the negation word is removed, and we get a sentence with inverted polarity as a result. Replacing with the antonym that has the maximum dissimilarity ensures that the maximum inversion of polarity takes place. This prevents words like "beautiful" being replaced by something like "bad", which is an indirect antonym.

Tokenization

Tokenization is the process of converting text into tokens before transforming it into vectors. It is also easier to filter out unnecessary tokens. For example, a document into paragraphs or sentences into words. In this case we are tokenising the tweets into words. At the same time, we use a RegexpTokenizer from NLTK to splits a string using a regular expression (regex) pattern that ignores all whitespace and punctuation.

Afterwards, we also run the data through a stop words filter provided also by NLTK. Stop words are the most commonly occurring words which are not relevant in the context of the data and do not contribute any deeper meaning to the phrase. By removing these stop words and the additional characters with the Regex Tokenizer, we can help to clean up each data entry.

Lemmatization

This process finds the base or dictionary form of each word known as the lemma and replaces each word in the input. This is done using vocabulary (dictionary importance of words) and morphological analysis (word structure and grammar relations). A lemmatizer is preferred over a stemmer in this case because it produces more readable words. Output that is easy to read is very desirable in topic modelling.

We use NLTK WordNet Lemmatizer which uses WordNet's built-in morphy function and returns the input word unchanged if it cannot be found in the WordNet.

Model Training

Performance Metrics used

In multi-class classification tasks such as cyberbullying detection, where each tweet is categorised into one of six classes ('age', 'ethnicity', 'gender', 'not_cyberbullying',

'other_cyberbullying', and 'religion'), a range of evaluation metrics is used to measure the model's performance across these specific categories.

- Precision measures the accuracy of positive predictions for each class. For instance, precision for the ethnicity class indicates the proportion of tweets predicted as ethnicity-based cyberbullying that were correct. High precision means fewer false positives in that class.
- Recall assesses how well the model identifies all instances of a particular class. For example, high recall in the gender class means the model successfully detects most tweets that involve gender-related bullying, with fewer false negatives.
- F1-score is the harmonic mean of precision and recall for each class, providing a single performance metric that balances both. It is especially useful when the class distribution is imbalanced, as it ensures neither false positives nor false negatives are overlooked.
- Support refers to the actual number of true instances for each class in the test set. For example, if the religion class has a support value of 1600, that means there were 1600 tweets truly belonging to that category, regardless of how the model predicted them.
- Accuracy is the overall proportion of correct predictions across all classes. While useful, it may be misleading in imbalanced datasets, as a model could achieve high accuracy by favouring the majority class (not_cyberbullying), while performing poorly on minority classes like age or religion.
- Macro average calculates the mean precision, recall, and F1-score by averaging across all six classes equally. This treats each class as equally important, regardless of how many samples it contains.
- Weighted average, on the other hand, also computes the mean precision, recall, and F1-score across classes, but considers the number of instances (support) per class. As such, it gives more weight to classes like not_cyberbullying if it has significantly more samples, reflecting a more representative view of overall performance in imbalanced datasets.

Together, these metrics offer a comprehensive view of model effectiveness, highlighting not just how often it is correct overall, but also how fairly and reliably it performs across different types of cyberbullying.

Pre-trained Transformers

The DeBERTa model was fine-tuned using a custom PyTorch training loop. The specific model used was microsoft/deberta-v3-base. Tokenizer used was Hugging Face AutoTokenizer with padding and truncation to 128 tokens.

The ELECTRA model was fine-tuned using Hugging Face's Trainer API. The specific model used was google/electra-base-discriminator. Tokenizer used was Hugging Face ElectraTokenizer, with padding and truncation up to 128 tokens.

The RoBERTa model was fine-tuned using Hugging Face's Trainer API. The specific model used was FacebookAI/roberta-base. Tokenizer used was Hugging Face RobertaTokenizer, with padding and truncation up to 128 tokens.

All the models using pre-trained transformers have the same configurations with the loss function as Cross Entropy Loss, optimizer as AdamW with a learning rate of $2e-5$ and 4 epochs of batch size 16.

The task is a multi-class classification problem, where each tweet belongs to one and only one cyberbullying category. Cross Entropy Loss is the standard and most effective choice for such problems, as it penalizes incorrect class predictions based on their predicted probabilities and encourages the model to confidently predict the correct class.

The AdamW optimizer is widely adopted in training transformer-based models due to its ability to handle sparse gradients and its built-in support for weight decay, which helps with generalization. A learning rate of $2e-5$ is commonly recommended for fine-tuning large pretrained language models like DeBERTa and ELECTRA. It offers a balance between training stability and convergence speed without overwhelming the pretrained weights with aggressive updates.

Based on empirical benchmarks and past research, 4 epochs is often sufficient for fine-tuning pretrained models without overfitting on moderately sized datasets.

A batch size of 16 was chosen to maintain a reasonable trade-off between convergence speed and memory usage, particularly for training on GPUs with limited VRAM. This batch size is also known to work well with the AdamW optimizer for transformer models.

Ensemble Methods

A Random Forest classifier was trained using a set of 25 features. These features aim to capture structural, lexical, sentiment-based, and syntactic patterns relevant to cyberbullying detection:

1. Structural Features

- `char_count`: Total number of characters in the text
- `word_count`: Total number of words
- `unique_word_count`: Number of unique words
- `avg_word_length`: Average length of words
- `punctuation_count`: Number of punctuation characters
- `capital_word_count`: Number of fully capitalized words
- `uppercase_ratio`: Ratio of uppercase letters to total characters
- `repeated_char_sequences`: Number of repeated character patterns (e.g., "soooo")

2. Lexical and Social Features

- `exclamation_count`: Number of exclamation marks
- `question_count`: Number of question marks
- `mention_count`: Number of @mentions
- `hashtag_count`: Number of hashtags
- `emoji_count`: Number of emojis
- `badword_count`: Count of words matching a profanity or slur lexicon
- `stopword_ratio`: Ratio of stopwords to total words

3. Sentiment Features

- `sentiment_polarity`: Polarity score from TextBlob sentiment analysis
- `sentiment_subjectivity`: Subjectivity score from TextBlob
- `vader_neg`: VADER negative sentiment score
- `vader_neu`: VADER neutral sentiment score
- `vader_pos`: VADER positive sentiment score
- `vader_compound`: VADER compound sentiment score

4. Part-of-Speech Features

- `noun_count`: Number of nouns in the text
- `verb_count`: Number of verbs
- `adj_count`: Number of adjectives
- `adv_count`: Number of adverbs

These features were chosen to balance interpretability and predictive power, enabling the Random Forest model to make decisions based on observable patterns in linguistic structure, emotional tone, and social cues.

The final classification system employed a stacking ensemble method, combining the outputs of three base models to improve overall predictive performance. These models included:

- A fine-tuned DeBERTa model
- A fine-tuned ELECTRA model
- A Random Forest model trained on engineered text features

The model optimizes log loss, which is appropriate for multi-class classification problems. It penalizes incorrect predictions based on the confidence of the predicted class probabilities.

The Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm was used as the default solver. It is efficient for convex optimization problems and performs well on moderate-sized datasets like the stacked feature set used here.

Testing

Results

DeBERTa – [Appendix A](#)

Classification	Precision	Recall	F1-score	Support
age	0.99	0.98	0.99	1598
ethnicity	0.98	0.97	0.98	1592
gender	0.86	0.92	0.89	1595
not_cyberbullying	0.74	0.61	0.67	1589
other_cyberbullying	0.71	0.79	0.75	1565
religion	0.96	0.96	0.96	1600
accuracy	-	-	0.87	9539
macro avg	0.87	0.87	0.87	9539
weighted avg	0.87	0.87	0.87	9539

The classification report in Appendix A summarizes the performance of the DeBERTa model across six categories of cyberbullying. The key insights are as follows:

- **Strong Overall Accuracy:** The model achieved an overall accuracy of 87%, with both the macro and weighted F1-scores also at 0.87, indicating consistent performance across all classes without overfitting to dominant categories.
- **High Precision in Specific Categories:** The model performed exceptionally well in detecting cyberbullying related to age, ethnicity, and religion, with F1-scores of 0.99, 0.98, and 0.96, respectively. This suggests high reliability when identifying structured and explicit forms of targeted abuse.
- **Lower Performance for 'Not Cyberbullying':** The 'not_cyberbullying' class had the lowest F1-score of 0.67, driven by a relatively low recall of 0.61. This indicates that the model tends to misclassify some neutral or benign messages as cyberbullying, which may increase false positives.
- **Moderate Results for 'Other Cyberbullying':** The F1-score for 'other_cyberbullying' was 0.75, highlighting the model's difficulty in correctly classifying more ambiguous or mixed-content bullying that doesn't fit neatly into predefined categories.

In summary, while DeBERTa excels in classifying clear-cut cases of cyberbullying, there is room for improvement in handling ambiguous cases and in reducing false positives for non-cyberbullying content.

ELECTRA – [Appendix B](#)

Classification	Precision	Recall	F1-score	Support
age	0.98	0.98	0.98	1627
ethnicity	0.99	0.99	0.99	1641
gender	0.89	0.90	0.90	1629
not_cyberbullying	0.70	0.61	0.65	1516
other_cyberbullying	0.68	0.77	0.72	1561
religion	0.96	0.97	0.97	1565
accuracy	-	-	0.87	9539
macro avg	0.87	0.87	0.87	9539
weighted avg	0.87	0.87	0.87	9539

The performance of the ELECTRA model, as shown in Appendix B, demonstrates strong overall classification capabilities, with an accuracy of 87% and matching macro and weighted F1-scores of 0.87 — identical to those achieved by the DeBERTa model. However, several distinctions in class-level performance are noteworthy:

Excellent Consistency in Core Classes:

- ELECTRA performs extremely well on structured categories such as age, ethnicity, and religion, with all three classes achieving F1-scores of 0.97 to 0.99. This is comparable to DeBERTa, though ELECTRA slightly outperforms it in ethnicity and gender classification (F1: 0.99 and 0.90, vs. DeBERTa's 0.98 and 0.89).

Better Handling of 'Gender':

- ELECTRA achieves an F1-score of 0.90 on the 'gender' class, outperforming DeBERTa's score of 0.89, suggesting improved recognition of gender-based cyberbullying.

Similar Weakness in 'Not Cyberbullying':

- Like DeBERTa, ELECTRA struggles with the 'not_cyberbullying' class, with an F1-score of 0.65 and a recall of 0.61. This indicates that both models have difficulty correctly identifying benign content, raising concerns about false positive classifications.
- Slightly Lower Precision for 'Other Cyberbullying': ELECTRA scores slightly lower in 'other_cyberbullying' with an F1-score of 0.72 compared to DeBERTa's 0.75, which may affect its ability to generalize to ambiguous or multi-faceted bullying content.

Overall, while ELECTRA and DeBERTa perform similarly on aggregate, ELECTRA shows slight gains in gender and ethnicity detection, whereas DeBERTa maintains better precision on ambiguous content like other cyberbullying. Choosing between them may depend on which types of errors are more acceptable in the application context.

RoBERTa – [Appendix C](#)

Classification	Precision	Recall	F1-score	Support
age	0.76	0.57	0.65	1624
ethnicity	0.70	0.78	0.74	1612
gender	0.99	0.97	0.98	1603
not_cyberbullying	0.96	0.96	0.96	1566
other_cyberbullying	0.80	0.94	0.87	1531
religion	0.98	0.98	0.98	1603
accuracy	-	-	0.87	9539
macro avg	0.87	0.87	0.86	9539
weighted avg	0.87	0.87	0.86	9539

As shown in Appendix C, the model achieved an overall accuracy of 87%, with a macro average F1-score of 0.87 and a weighted F1-score of 0.86. While this performance is comparable to both DeBERTa and ELECTRA at an aggregate level, there are notable distinctions in class-specific performance.

High performance in core classes:

- The model demonstrated exceptional capability in detecting gender, not_cyberbullying, and religion, achieving F1-scores of 0.98, 0.96, and 0.98 respectively. These results indicate strong precision and recall, suggesting that RoBERTa is effective in identifying both harmful and non-harmful content when the patterns are well defined.

Superior handling of ambiguous cases:

- RoBERTa achieved an F1-score of 0.87 for the other_cyberbullying class, outperforming all other individual models. This category typically includes diverse and less explicitly defined forms of abuse, highlighting RoBERTa's robustness in handling nuanced language.

Lower recall for the age category:

- Despite solid precision, the model's recall for age-related bullying was only 0.57, resulting in an F1-score of 0.65. This indicates that many true instances of age-related abuse were missed, potentially due to subtler language patterns in this category.

While RoBERTa matched DeBERTa and ELECTRA in terms of overall accuracy, it stood out in its ability to accurately detect subtle and ambiguous forms of cyberbullying, especially in the other_cyberbullying and not_cyberbullying categories. However, its relatively weaker performance in identifying age-related abuse highlights an area for further tuning.

Random Forest – [Appendix D](#)

Classification	Precision	Recall	F1-score	Support
age	0.67	0.69	0.68	1598
ethnicity	0.83	0.66	0.74	1592
gender	0.66	0.60	0.63	1590
not_cyberbullying	0.45	0.50	0.47	1587
other_cyberbullying	0.46	0.52	0.49	1565
religion	0.65	0.66	0.65	1600
accuracy	-	-	0.61	9532

macro avg	0.62	0.61	0.61	9532
weighted avg	0.62	0.61	0.61	9532

Despite the richness of the engineered features, the model achieved a modest overall accuracy of 61%, with both macro and weighted F1-scores also at 0.61. This suggests that while the features capture surface-level and emotional traits of the text, they lack the deep semantic understanding needed for precise classification across nuanced categories of cyberbullying.

- **Best Performing Class:** The ethnicity category showed the strongest performance, with an F1-score of 0.74, likely due to identifiable patterns in language or keyword cues (e.g., specific slurs or phrases) captured by the badword_count, sentiment, or noun/adj features.
- **Mid-Level Performance:** Other categories such as religion, gender, and age had F1-scores ranging from 0.63 to 0.68, indicating moderate predictive power for more structured forms of abuse where stylistic features may partially reflect the class.
- **Weakest Classes:** The model struggled most with 'not cyberbullying' and 'other cyberbullying', with F1-scores of 0.47 and 0.49 respectively. This reflects its difficulty in distinguishing benign content from subtle or ambiguous bullying when only shallow features are used. These classes likely require deeper contextual and semantic understanding beyond what hand-crafted features can provide.

While the Random Forest model offers a lightweight and interpretable alternative to transformer-based models, its performance is significantly lower across all categories. The results underscore the limitations of relying solely on engineered surface features for a complex language task like cyberbullying classification. This model may still be useful in scenarios with constrained resources or as a base learner in ensemble setups like stacking, but on its own, it lacks the accuracy required for high-stakes applications.

Stacking (DeBERTa, Electra, Random Forest) – [Appendix E](#)

Classification	Precision	Recall	F1-score	Support
age	0.99	0.99	0.99	1598
ethnicity	0.99	0.97	0.98	1592
gender	0.93	0.95	0.94	1595
not_cyberbullying	0.66	0.81	0.73	1589
other_cyberbullying	0.75	0.58	0.66	1565
religion	0.98	0.98	0.98	1600

accuracy	-	-	0.88	9539
macro avg	0.89	0.88	0.88	9539
weighted avg	0.89	0.88	0.88	9539

The stacking ensemble combines the strengths of three base models — DeBERTa, ELECTRA, and a Random Forest classifier trained on engineered features. As shown in Appendix E, this stacked model achieves the highest overall performance among all evaluated models:

- Overall Accuracy: 0.88
- Macro and Weighted F1-scores: 0.88

This represents a 1% improvement over DeBERTa and ELECTRA, which both plateaued at 0.87 accuracy and macro F1-score. While the improvement may appear incremental, it reflects better class-level consistency, particularly for the more difficult categories.

Top-performing Classes:

- ‘age’, ‘ethnicity’, and ‘religion’ maintain near-perfect F1-scores (0.98–0.99), consistent with both DeBERTa and ELECTRA, showing that stacking retains the strengths of its transformer components.
- ‘gender’ reaches an F1-score of 0.94, improving upon both DeBERTa (0.89) and ELECTRA (0.90), indicating more reliable recognition of gender-based abuse.

Improved 'Not Cyberbullying' Classification:

- F1-score increases to 0.73, a substantial improvement over DeBERTa (0.67), ELECTRA (0.65), and especially Random Forest (0.47).
- This suggests that stacking allows better differentiation between harmful and non-harmful content, likely due to the diversity of input perspectives across models.

Balanced Trade-off for 'other_cyberbullying':

- F1-score of 0.66 is higher than Random Forest (0.49) and comparable to DeBERTa (0.75) and ELECTRA (0.72).
- Precision is high (0.75) but recall drops to 0.58, indicating a tendency toward conservative classification — fewer false positives, but possibly more false negatives.

The stacking model offers the best overall performance across nearly all metrics, combining the semantic depth of transformer models with the feature-driven precision of Random Forest. It particularly improves robustness in challenging classes like 'not_cyberbullying' and 'gender', making it a strong candidate for deployment in real-world cyberbullying detection systems where both accuracy and reliability across categories are critical.

Comparison against Baseline Model

We compare the model against the baseline (Google Jigsaw API). We apply for access using Google Cloud Console and receive an API key which we can use to access the API. However, the free tier only allows for 60 requests a minute, so we only test 500 tweets from the dataset (250 cyberbullying and 250 not cyberbullying).

We record the toxicity score returned by the API (0.00 to 1.00), then, we classify the tweets as toxic or not, experimenting with different thresholds:

```
thresholds = [i / 100 for i in range(10, 91)]
```

The best threshold is 0.18 with an F1 score of 0.7617.

The reason for this low score is twofold:

1. The Jigsaw API is not trained to detect cyberbullying specifically, only toxicity.
2. Some tweets in the dataset are mislabelled, especially in the not_cyberbullying class. For example: “@QolooB4 @Vandaliser @sajid_fairooz @IsraeliRegime I don't care about an idiot with an absurd opinion” is tagged not_cyberbullying.

Challenges

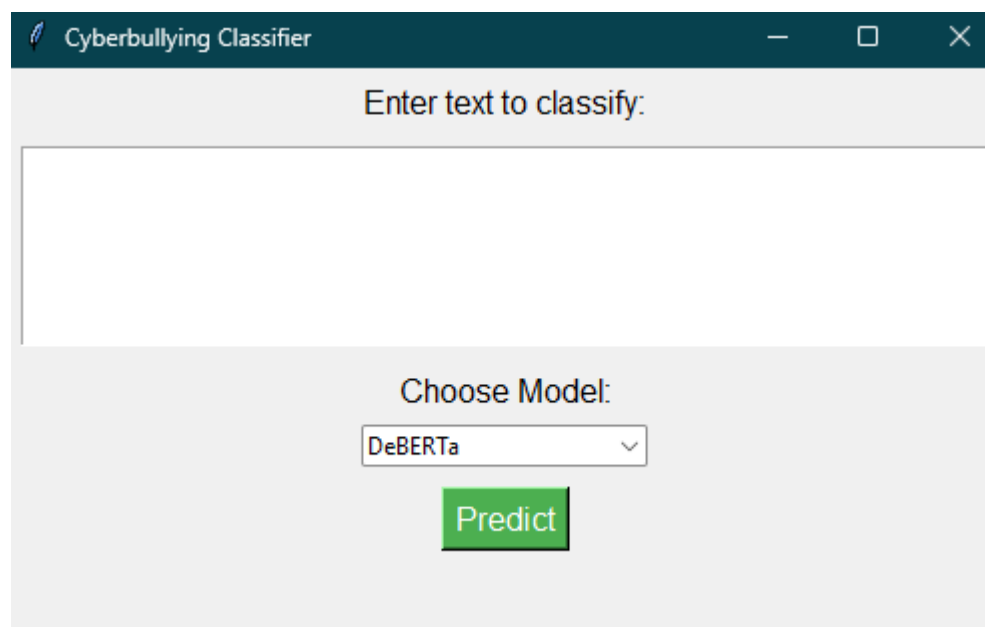
Detecting cyberbullying through AI presents several critical challenges that impact both the development and deployment of effective models.

First, the inherent ambiguity of online content, particularly on platforms like X (formerly Twitter), poses a significant obstacle. Cyberbullying often manifests through subtle language cues, sarcasm, irony, or cultural references, which are difficult for models to interpret accurately. This is especially evident in the misclassification of tweets within the "other_cyberbullying" and "not_cyberbullying" categories, where context is key and even state-of-the-art transformer-based models struggle to discern intent, resulting in a higher rate of false positives and negatives.

Second, while ensemble models that integrate advanced transformer architectures such as ELECTRA and DeBERTa demonstrate improved classification performance, they also demand considerable computational resources. The increased complexity and training overhead make real-time moderation or deployment on low-resource environments less feasible, posing practical limitations for widespread adoption.

Lastly, despite relying on a balanced and manually annotated dataset, inconsistencies in labelling remain a concern. Variability in human judgement and occasional annotation errors, particularly in borderline or nuanced cases, can compromise the reliability of training data. These issues underline the necessity for more rigorous annotation standards and enhanced quality control measures to ensure the generalisability and robustness of cyberbullying detection systems.

Graphical User Interface



The graphical user interface for this project was developed using Tkinter, Python's standard library for creating simple GUIs. Tkinter was chosen due to its ease of use, lightweight nature, and tight integration with Python — making it ideal for building a quick and functional interface without requiring external dependencies.

The GUI is implemented in a Python script named `GUI.py`. To run it, users simply need to execute the script in a Python environment with the required dependencies installed. Upon launching, the interface presents the following elements:

- A text box where users can enter or paste any message they wish to classify.
- A model dropdown selection menu that allows users to choose between multiple classification models
- A "Predict" button that triggers the classification.

Once the button is clicked, the chosen model processes the input and returns the classification result. This output is displayed within the interface, indicating whether the text contains cyberbullying, and if so, it specifies the type of cyberbullying detected (e.g., age, ethnicity, gender).

Conclusion

This project successfully demonstrates the effectiveness of using ensemble-based machine learning models to detect and classify cyberbullying on social media, particularly on platforms like X (formerly Twitter), where harmful speech is both prevalent and complex. By leveraging the complementary strengths of transformer-based models, ELECTRA, DeBERTa, RoBERTa, and traditional methods such as Random Forest, we developed a robust system capable of nuanced sentiment analysis and multi-class classification across six categories of cyberbullying.

The stacked ensemble model emerged as the best-performing architecture, achieving an overall accuracy of 88% and significantly improving classification performance, especially for ambiguous and often misclassified categories like “not_cyberbullying” and “gender.” This model not only outperformed individual models but also surpassed the baseline Google Jigsaw API in both accuracy and contextual understanding, proving the advantage of task-specific training.

In addition to strong model performance, we implemented a lightweight and user-friendly GUI to allow easy interaction with the models, making the system accessible for practical applications such as content moderation and research.

Nevertheless, challenges remain. The ambiguous nature of cyberbullying, the computational demands of large transformer models, and annotation inconsistencies within datasets all underscore the need for continued development. Future work may involve refining training data through more rigorous annotation, optimizing models for deployment in low-resource settings, and enhancing interpretability to support human moderators.

In summary, this project contributes a scalable and accurate solution for cyberbullying detection, aligned with the broader goal of promoting safer digital environments through AI for Good.

Peer Review

For our group, the work was split evenly across all members, with each person contributing equally to code, report, and presentation tasks. Therefore, all members deserve 100% and the full 4 marks.

Name	Peer Review / Work Allocation
Lee Chien Shyong	100%
John-David Tan Ming Sheng	100%
Zachary Seow Shijie	100%
Noven Zen Hong Guan	100%
Genson Low Yuan Sen	100%

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Appendix

Appendix A - DeBERTa Results

	precision	recall	f1-score	support
age	0.99	0.98	0.99	1598
ethnicity	0.98	0.97	0.98	1592
gender	0.86	0.92	0.89	1595
not_cyberbullying	0.74	0.61	0.67	1589
other_cyberbullying	0.71	0.79	0.75	1565
religion	0.96	0.96	0.96	1600
accuracy			0.87	9539
macro avg	0.87	0.87	0.87	9539
weighted avg	0.87	0.87	0.87	9539

Appendix B – ELECTRA Results

	precision	recall	f1-score	support
age	0.98	0.98	0.98	1627
ethnicity	0.99	0.99	0.99	1641
gender	0.89	0.90	0.90	1629
not_cyberbullying	0.70	0.61	0.65	1516
other_cyberbullying	0.68	0.77	0.72	1561
religion	0.96	0.97	0.97	1565
accuracy			0.87	9539
macro avg	0.87	0.87	0.87	9539
weighted avg	0.87	0.87	0.87	9539

Appendix C - RoBERTa

	precision	recall	f1-score	support
not_cyberbullying	0.76	0.57	0.65	1624
other_cyberbullying	0.70	0.78	0.74	1612
age	0.99	0.97	0.98	1603
religion	0.96	0.96	0.96	1566
gender	0.80	0.94	0.87	1531
ethnicity	0.98	0.98	0.98	1603
accuracy			0.87	9539
macro avg	0.87	0.87	0.86	9539
weighted avg	0.87	0.87	0.86	9539

Appendix D – Random Forest

	precision	recall	f1-score	support
age	0.67	0.69	0.68	1598
ethnicity	0.83	0.66	0.74	1592
gender	0.66	0.60	0.63	1590
not_cyberbullying	0.45	0.50	0.47	1587
other_cyberbullying	0.46	0.52	0.49	1565
religion	0.65	0.66	0.65	1600
accuracy			0.61	9532
macro avg	0.62	0.61	0.61	9532
weighted avg	0.62	0.61	0.61	9532

Appendix E – Stacking

	precision	recall	f1-score	support
age	0.99	0.99	0.99	1598
ethnicity	0.99	0.97	0.98	1592
gender	0.93	0.95	0.94	1595
not_cyberbullying	0.66	0.81	0.73	1589
other_cyberbullying	0.75	0.58	0.66	1565
religion	0.98	0.98	0.98	1600
accuracy			0.88	9539
macro avg	0.89	0.88	0.88	9539
weighted avg	0.89	0.88	0.88	9539